Lesson 4 MoveIt Introduction

1. MoveIt Overview

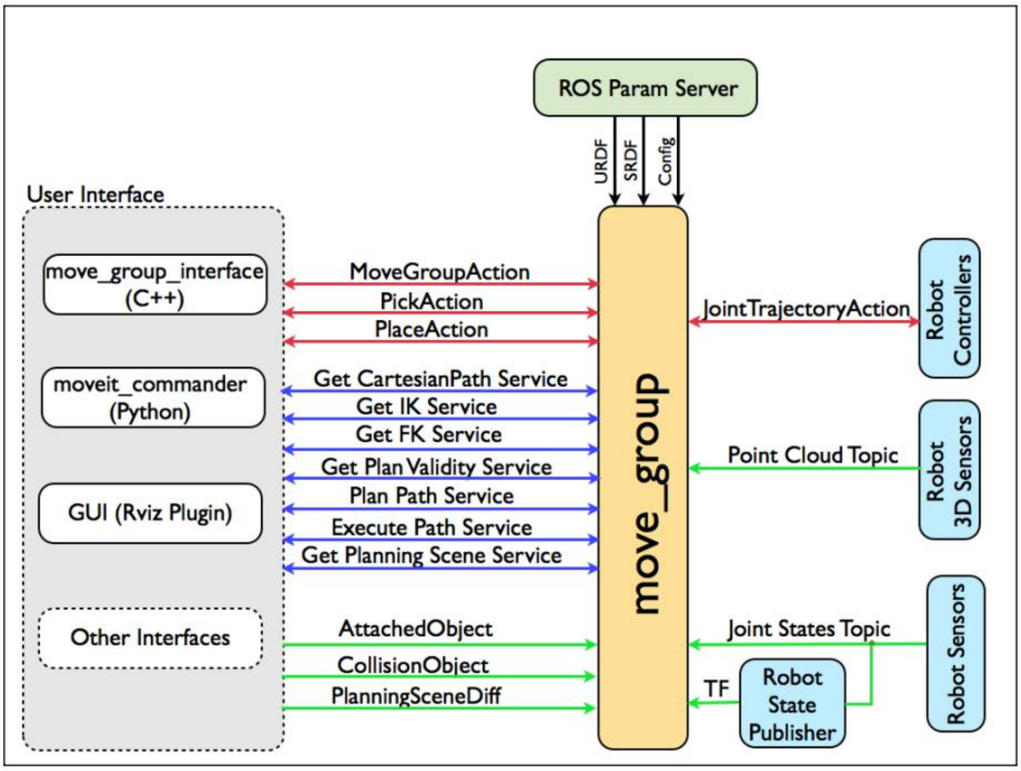
As the most advanced software for mobile operation, MoveIt consists of a series of mobile operating feature packs based on Ros system, including motion planning, operation control, 3D perception, kinematics, collision detection, etc.

MoveIt provides an user-friendly platform, which is used to develop advanced robot application and evaluate new robot design as well as building integrated robotic product. Being the most widely-used open source operation software, it is applied in industry, business, research and development and other field.

In addition, MoveIt provides a series of mature plug-ins and tools to realize quick configuration of robotic arm control. What's more, thanks to its multiple API, it is convenient for users to perform secondary development on MoveIt to develop more creative application.

2. MoveIt System Structure

MoveIt system structure is as follow:



MoveIt system structure comprises of four parts, namely User Interface, move_group, ROS Param Server and external robot parts.

1) User Interface: User can access the operation and service offered by move_group through one of the following three methods

● C++: move_group_interface package makes it easy for you to use move_group

● Python: it takes advantage of MoveIt_commander package

● GUI: it utilizes Rviz of Motion Planning

2) move_group: Ros node. It obtains three kinds of messages through ROS param server

● URDF: move_group searches for robot_description parameter on ROS parameter server to obtain the URDF of robot.

● SRDF: move_group searches for robot_description_semantic parameter on ROS parameter server to obtain the SRDF of robot.

- MoveIt configuration: move_group seeks for other specific MoveIt configurations on ROS parameter server, including joint limits, kinematics, motion plan, perception and other information.

3) ROS Param Server: It provides ROS node with URDF, SRDF, MoveIt configuration information.

4) External robot parts: Robot Sensors, Robot Sensors and Robot Controllers.

3. Plug-in of MoveIt--RViz

3.1 RViz Introduction

RViz is three-dimensional visual platform of ROS system and the plug-in of MoveIt, which can visualize the external information and send control information to monitored object.

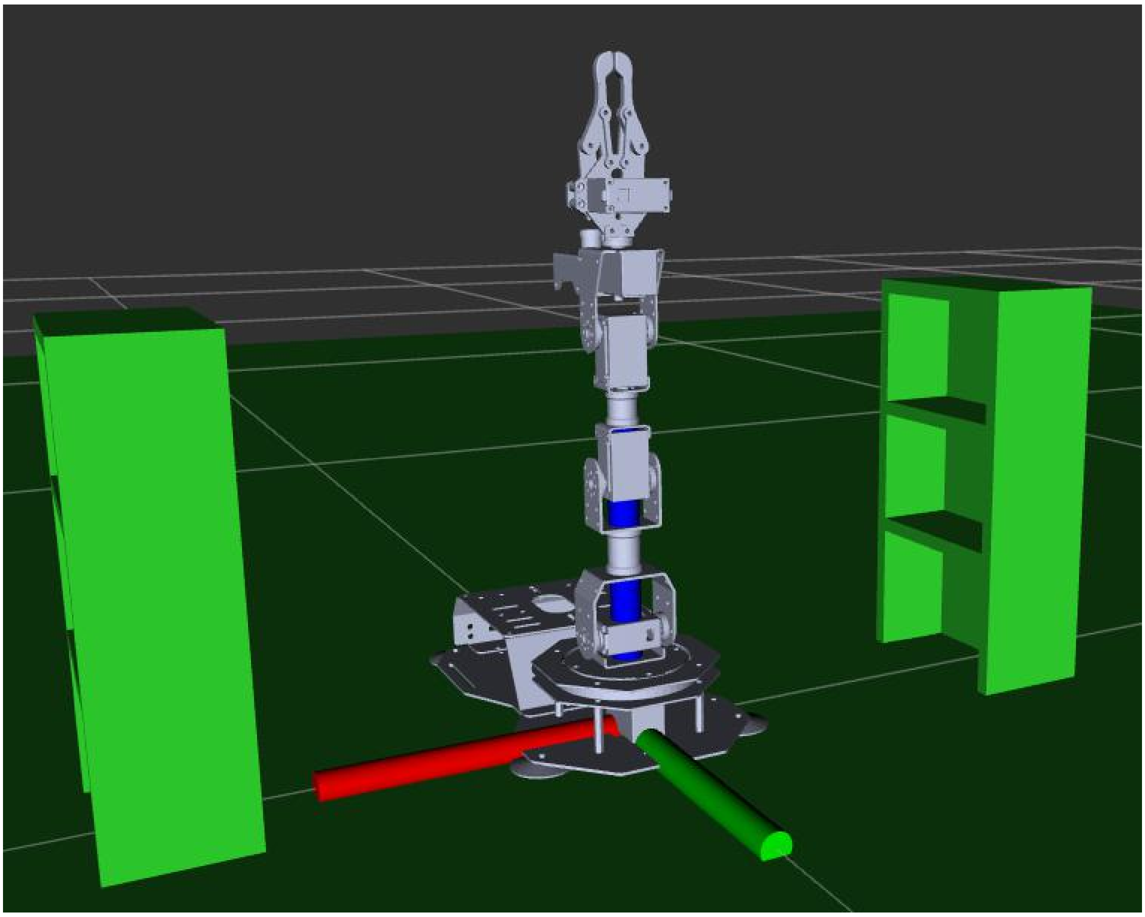
Through Rviz, user can set virtual environment, set the robot's start status and target status in interaction mode, and test various motion programming algorithm for visual output

3.2 RViz Coordinate System

Coordinate system is essential for forward and inverse kinematics analysis. We will introduce some common coordinates taking the example of ArmPi FPV robotic arm in RViz environment.

1) World Coordinate System

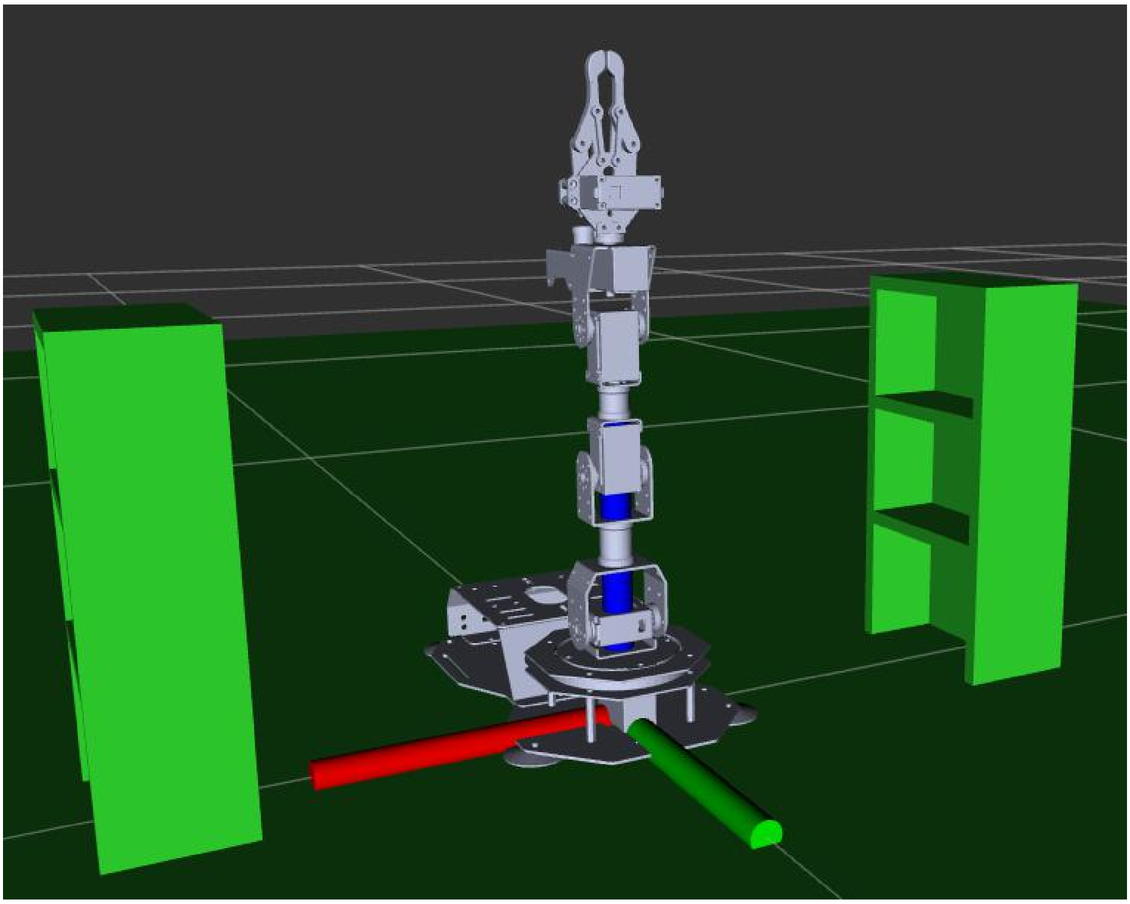
The World Coordinate System is a stationary reference Coordinate System located on the “ground”. It doesn't translate or rotate over time. And, the "ground" here is in a broad sense. For example, when ArmPi FPV robotic arm is placed on the desktop, the world coordinate system will be on the plane of the table.



2) Robot Arm Base Coordinate System

Base Coordinate System is the short form of Robot Arm Base Coordinate System, locating on the base of robotic arm. If there is only one robotic arm, the robot arm base coordinate system will overlap world coordinate system in simple application scenarios.

The base coordinate system will make right-hand rule, with bottom servo as origin, red axis as X axis, green axis as Y axis and blue as Z axis.

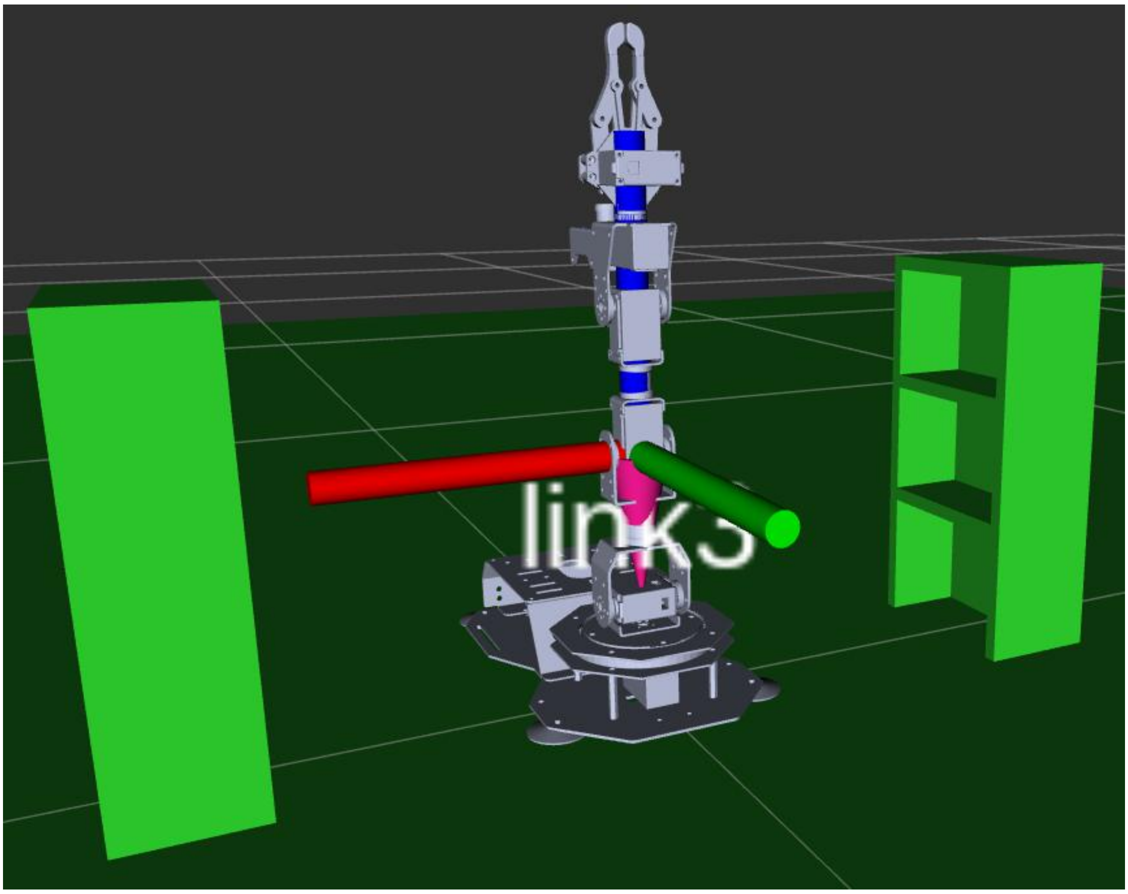


3) Joint Coordinate System

The rotating joint and the linkage are the important part of robotic arm body. And Joint Coordinate System is a Coordinate System built on Joint.

And, Joint Coordinate System also follows the right-hand-rule, with pink cone as origin, red axis as X axis, green axis as Y axis and blue as Z axis.

The number of joint coordinates is not fixed, which is related to the number of joints. So, you have to be very careful when you set up the joint coordinate system.

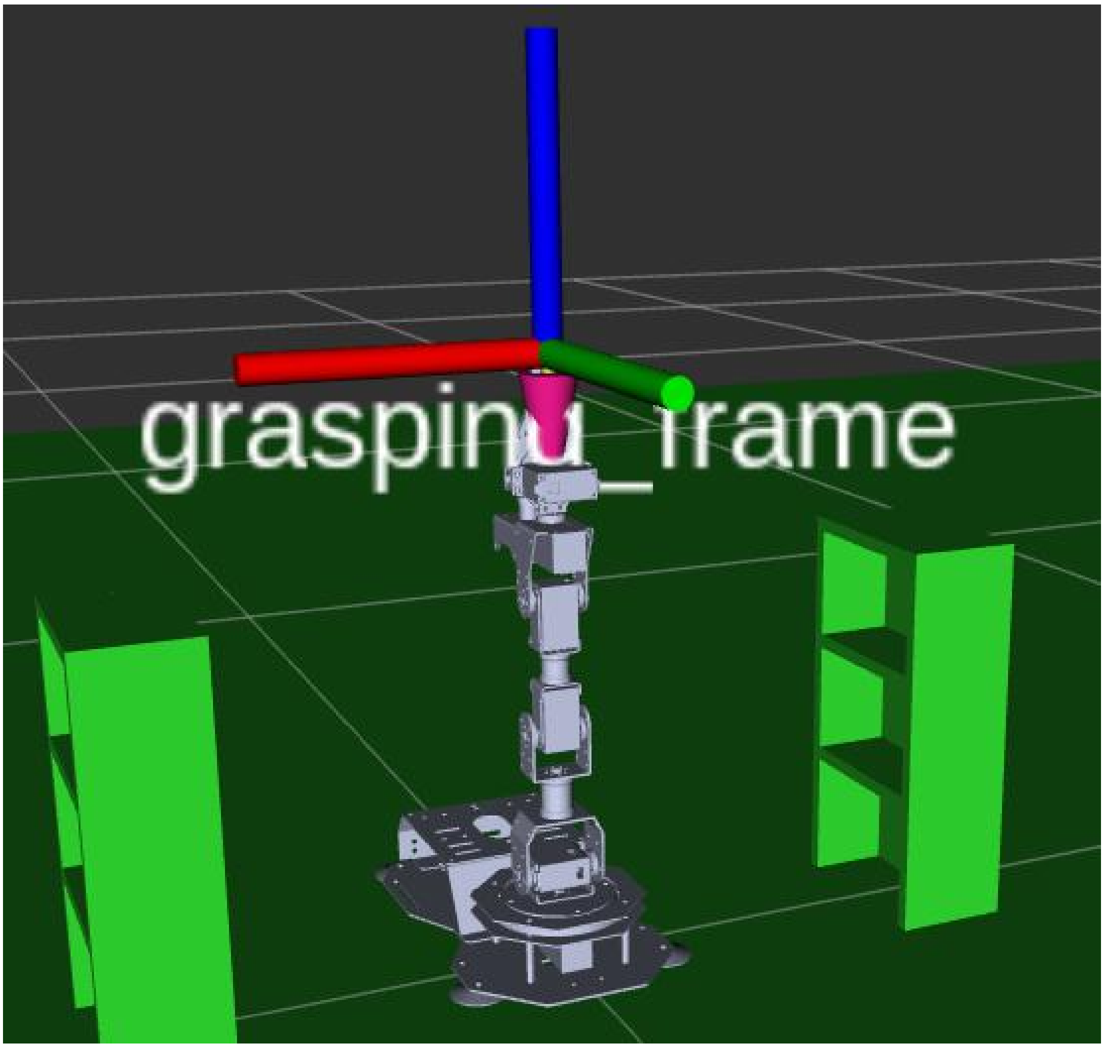


4) Tool Coordinate System

We will install the corresponding tool on the end of robotic arm to meet various operation needs. Let us take the gripper for example.

Tool Coordinate System usually takes Tool Center Point as the origin and follow the right-hand-rule to build the coordinate system. This method of defining the origin is suitable for the gripper-typed tool, and if the tool itself is more complex, it can be divided into several coordinates.

In the model below , the tool coordinate system of the gripper takes the pink cone as origin, red axis as X axis, green axis as Y axis and blue axis as Z axis.



Please move to our website, https://MoveIt.ros.org/documentation/concepts/, for more information about MoveIt!